

# Bohr Flop in C minor

Lead sheet · pitch read from a resonant driven qubit · partner voice is the singlet anti-correlation · quarter = 108

The image displays a musical score for a piece titled "Bohr Flop in C minor". The score is written for two systems, each consisting of a treble and bass staff. The key signature is C minor (three flats: Bb, Eb, Ab) and the time signature is 4/4. The tempo is indicated as "quarter = 108".

The notation includes:

- Melody P(e):** The upper line of each system, written in treble clef, representing the excited-state probability of H = (Omega/2) sigma-x, sampled twice a second.
- Partner 1 - P(e):** The lower line of each system, written in bass clef, representing the paired outcome, 1 - P.

The score features a series of eighth notes and rests, with vertical lines connecting the two staves in each system, indicating the singlet anti-correlation between the partner voice and the melody.

C minor. Eighth notes. Upper line: excited-state probability of H = (Omega/2) sigma-x, sampled twice a second. Lower line, bass clef: the paired outcome, 1 - P.

# What is quantum in this piece

A speaker only moves air. The quantum part is which numbers were allowed to become frequencies.

- **The oscillator**

A quantum harmonic oscillator in the equal superposition  $(|0\rangle + |1\rangle + |2\rangle) / \sqrt{3}$  has energies  $E(n) = \hbar \omega (n + 1/2)$ . The only tones in its time evolution are the Bohr differences,  $\omega$  and  $2\omega$ . Those are the 220 Hz drone and the 440 Hz partial on the recording. Equal spacing is why the oscillator is called harmonic.

- **The qubit**

A resonant driven qubit, Hamiltonian  $H = (\Omega/2) \sigma_x$ , starts in  $|0\rangle$ . The chance it is found excited is  $P(t) = \sin^2(\Omega t / 2)$ , with one full flop about every two seconds. That probability is quantized onto C minor and becomes the upper voice. In the audio, notes dwell longer near the poles, where the state sits at  $|0\rangle$  or  $|1\rangle$ . The printed score evens them to eighths so the line can be read.

- **The partner voice**

The lower staff is not a harmony written by ear. It reads  $1 - P$ , the anti-correlation of a spin singlet: if one outcome is up, the paired outcome is down. It is written in bass clef so the paired register sits on the staff.

- **What it is not**

Nothing in the file is in superposition after it leaves the calculation. The PDF, the wav, and a browser playback are ordinary. The score is a map, and the map is exact enough to recompute.

## ABC, for MuseScore or Flat

X:1 T:Bohr Flop in C minor M:4/4 L:1/8 Q:1/4=108 K:Cmin

\_E, \_B F C, \_B, c \_A, D, G \_A \_E, F, \_B \_E C, D |

Melody only. The partner is the index-reversed scale degree, written an octave lower in bass clef.